Reverse Engineering for Additive Manufacturing 242

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Class Outline

Reverse Engineering
Reverse Engineering in Additive Manufacturing
The Basic AM Process
Reverse Engineering Applications
Scanning Methods for RE
RE for AM Review
Point Cloud and Mesh Data
The Reverse Engineering Process
Reverse Engineering for AM Part Inspection
Final Review

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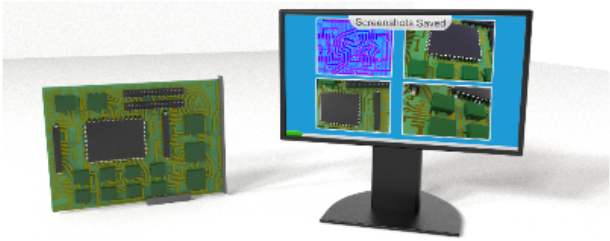
Lesson: 1/10

Reverse Engineering



Engineering is the field of designing and creating machines, electronics, buildings, parts, and structures. For example, engineering creates both roads and the vehicles that travel them. Reverse engineering (RE), in contrast, is the process of determining the design and creation of an existing object. Reverse engineering, also known as back engineering, has a variety of purposes and is used in multiple industries. Computer hardware manufacturers, for example, often reverse-engineer microchips by disassembling them and saving photographs of each layer. These photos form a record of the chip components and how they are assembled, which engineers can then refer to if needed. Similarly, mechanical engineers can use RE to determine how a vintage car or truck engine was manufactured. Part designers often use digital scanning tools and computer-aided design (CAD) software to reverse-engineer and recreate existing parts. These RE processes may be used to keep a library of part designs, create a replica of a rare or vintage part, or analyze a part to improve on the next version.

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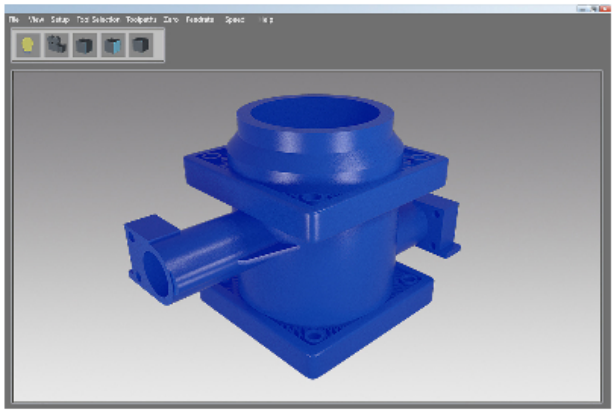
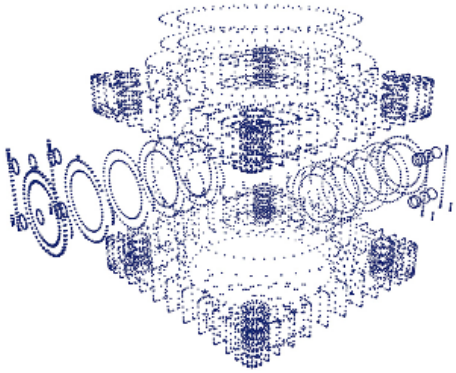
Lesson: 2/10

Reverse Engineering in Additive Manufacturing



CAD Model

The AM design and engineering process generally consists of designing a part, creating a digital model of the part in CAD software, and converting the CAD file into usable machine instructions so the AM machine can build the part. Consequently, reverse engineering a part in order to recreate it using AM requires the creation of a precise digital CAD model, generally through scanning the part.



Point Cloud

Scanning tools, such as 3D laser light scanners, can record the geometry, or shape, of a part and replicate it as a digital point cloud. A point cloud is a grouping of virtual dots that represent locations on a part's surface. Connecting these dots forms a digital image of the part's surface called a mesh. A mesh is similar to a point cloud, but it connects points to form an image using interconnected shapes. Engineers can then convert the mesh to a CAD model in order to build the part with an additive process.

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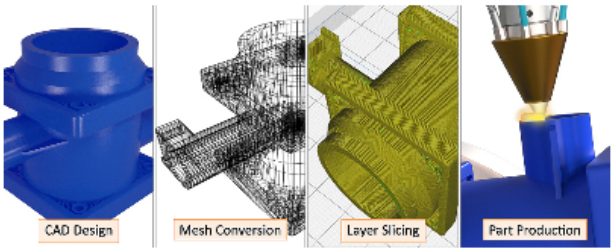
The Basic AM Process



Like any part production process, AM begins with part design. An engineer or part designer generally begins by creating a three-dimensional (3D) virtual model in CAD software. CAD software allows designers to not only see what a part will look like, but to simulate how it will react to various conditions and uses. The CAD model must then be converted to an STL or AMF file. STL is a type of mesh software that uses interconnected triangles to represent the part surface. AMF is a newer format that functions similarly but also provides more advanced data, such a colors. Since AM processes deposit or fuse material in layers, it is generally then necessary to use a slicer software to divide the model into the layers that the AM machine will deposit. An engineer or specialized software then creates a G code program or a machine-specific build file to direct the movements of the AM machine during the part production process. Reverse engineering begins with scanning a physical part in order to create a virtual model of the part. Once the part exists as a digital model, engineers can begin the AM part design and production process.

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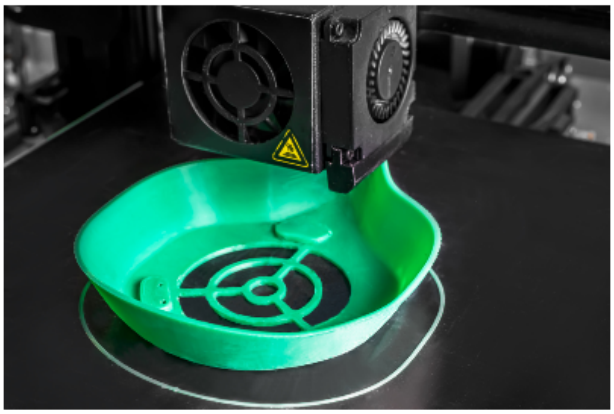
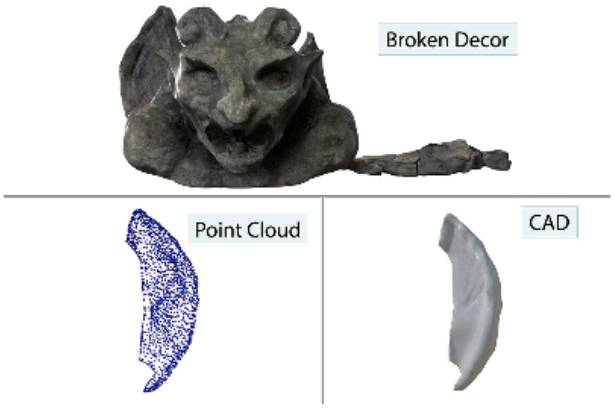
Reverse Engineering Applications

RE with AM can be incredibly helpful in areas that are far from cities or manufacturing facilities. Employees on remote worksites, including military personnel, can reverse-engineer parts to repair or maintain equipment. Using RE and AM in the field often takes much less time and costs less than shipping the part.



Similarly, medical personnel in areas with limited access to supplies can also benefit from reverse engineering and AM. RE with AM can replicate medical equipment and devices such as respirators or even prosthetics. In response to the COVID-19 pandemic, some manufacturers used RE and AM to respond to critical shortages and create respirators for healthcare workers. Engineers have even used RE to additively manufacture a prosthetic elbow joint that functions almost identically to a biological human elbow.



RE can also help to preserve and restore historic and cultural structures. If, for example, a historic building has crumbling concrete decorations, engineers can scan them to create a point cloud. If the decoration has broken or missing areas, the point cloud will record them.

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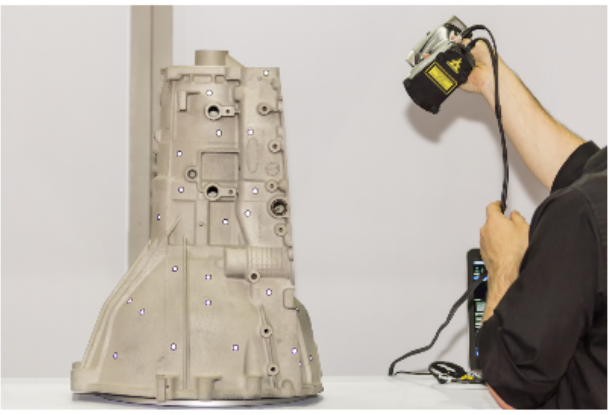
Once the point cloud is converted to a CAD file, engineers can manipulate the 3D model to fill in any missing or flawed areas. They can also use binder jetting, material extrusion, or another process to rebuild the concrete item.



In addition, RE can be useful when designing new products. Reverse engineering a newly designed part or prototype can help designers to find flaws or issues that they may have overlooked during the design process. Similarly, reverse engineering an existing part can help designers improve upon its design and increase the quality of future products.

Unfortunately, there are also illegal uses for reverse engineering. In the United States and some other countries, part and product designs are considered proprietary to the company that created them. This means that companies own their part designs, and it is theft to use their design without permission. If someone reverse engineers a proprietary product and recreates it, the person can face criminal penalties.

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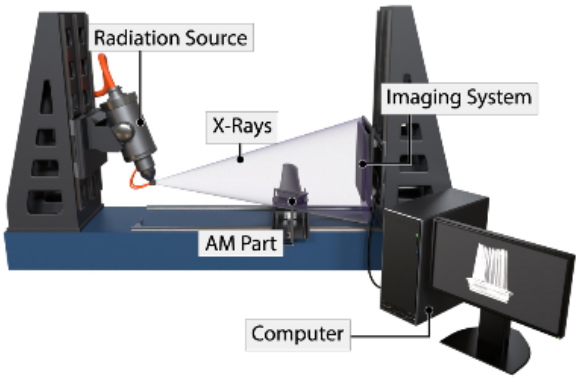
Scanning Methods for RE

Camera Drone Controller

Photogrammetry utilizes photography to create images of an object from multiple angles. Rather than creating a point cloud or mesh, photogrammetry software uses photos as data to construct a 3D model of an object. It generally requires at least 100 photos for the software to construct an accurate model of an object's geometry. Since photogrammetry requires only cameras and photogrammetry software, the initial cost is lower than other RE scanning methods. It is also an effective way to scan large objects or areas. Engineers can attach cameras to a drone, which can fly over tall buildings or other large objects to scan areas that would otherwise be impossible to reach. However, photogrammetry is much less accurate than other methods.



Computed Tomography Scanning System

Computed tomography (CT) scanning is another method used in reverse engineering with AM. CT scanning is similar to the CAT scans used in the medical field. It uses X-rays, which create an image of an object's interior by recording the amount of radiation that passes through it. Instead of creating just one image, CT scanning rotates the item to create multiple images from many different angles and locations. A digital imaging system connected to a computer records and collects these images. Similar to photogrammetry, specialized computer software then reconstructs the images into a 3D model of the object. However, CT scanning is far more precise and accurate than photogrammetry since it scans the part's internal structure.

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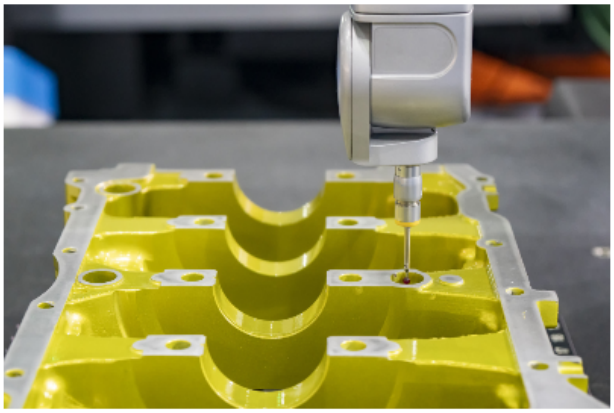
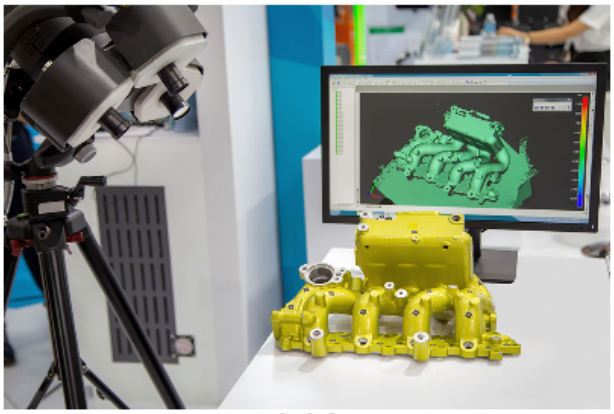


Coordinate Measuring Machine

Engineers can also use a coordinate measuring machine (CMM) to capture an object's geometry. CMMs are metrology tools that work by physically contacting multiple points of a part or item with a sensitive stylus. Traditional CMMs can be used only on objects small enough to fit within the machine's rigid frame. However, portable CMMs such as CMM arms are now available for objects that are larger or cannot be moved. A CMM creates a point cloud based on the data it collects, so it is crucial to use enough points of contact to create an accurate digital model.

Laser Scanner

Laser scanning is a type of 3D optical scanning that is also used in RE and metrology. Laser scanners are generally portable or hand-held devices that project a laser onto an object while a camera captures its reflection. Lasers are highly concentrated beams of light that bend or distort when they meet a surface. The amount and direction of distortion indicates a part's shape and location in physical space. Software then compiles the laser reflection data into a 3D model of the object. Laser scanners can use one or more lasers to detect different geometries, and they are very accurate and precise. Laser scanners can also be mounted on CMMs for noncontact scanning.



Structured Light Scanner

Structured-light scanning, like laser scanning, is a type of 3D optical scanning. Rather than a laser, structured light scanners project a pattern of light onto an object. A camera captures the distortion of the light pattern as it contacts the object, and software compiles the reflection data into a digital 3D model. Structured light scanners are mounted rather than hand-held, which increases their accuracy, and they can also cover more area of an object with each scan than a laser scanner.

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Lesson: 6/10

RE for AM Review

1. It is possible to use reverse engineering for criminal purposes. **TRUE**
2. Laser scanners for RE are too sensitive to be mounted on a coordinate measuring machine. **FALSE**
3. It is possible to reverse-engineer and replace human body parts for medical applications. **TRUE**
4. CAD software creates toolpaths that control tool movements on AM machines. **FALSE**
5. CT scanning creates images using radiation. **TRUE**
6. A point cloud consists of many connected triangles, similar to STL files used in AM. **FALSE**

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Point Cloud and Mesh Data

Each point in a point cloud represents a location in physical space. The point locations are defined using the Cartesian coordinate system, which consists of three straight lines, or axes. In the Cartesian coordinate system, the X axis is horizontal in a left-to-right direction, the Y axis is horizontal in a front-to-back direction, and the Z axis is vertical. The point where all three axes meet is called the origin. Any point in physical space can be located and defined by its position relative to the axes and origin.

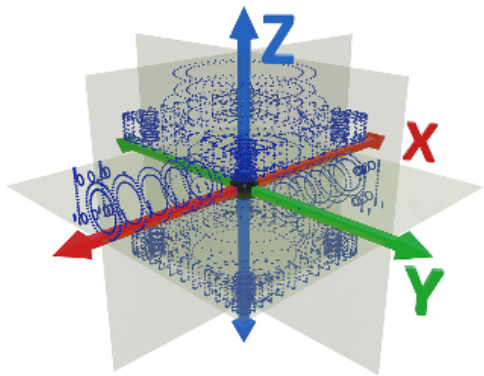

Figure 1: Cartesian Coordinate System

Once an engineer has scanned their object and created a point cloud, they use point cloud modeling software to convert it into a mesh. A mesh essentially connects the points in a point cloud with lines. This creates a number of connected two-dimensional (2D) shapes called polygons, which give the appearance of a mesh or net.

An important function of point cloud modeling software is cleaning up the point cloud. Since laser and structured-light scanners are such precise, sensitive instruments, even small variations, flaws, dirt, or particles on an object's surface can add points to the cloud. These stray points are known as noise, or scatter. Noise can distort the mesh and cause it to be less accurate, so it should be removed. In addition to cleaning and converting the cloud, engineers can also use point cloud modeling software to modify the mesh to fill in broken areas of a scanned part or make other adjustments. The mesh can then be imported into CAD software to create the final model.

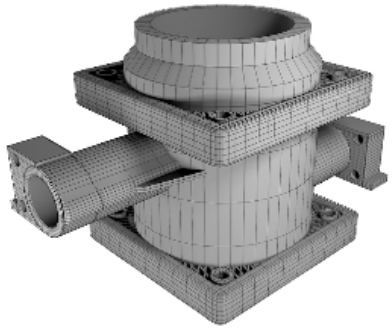

Figure 2: Mesh

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The Reverse Engineering Process
A manufacturer that creates molds and molded parts is incorporating additive manufacturing into their workflows. They want to shift from traditional, straight cooling channels to far more effective conformal cooling channels. In order to achieve conformal cooling, they plan to reverse engineer existing molds and modify the cooling channels.

First, an engineer uses a laser scanner to scan the mold and create a point cloud. It is important that the mold is clean and free of debris to reduce noise in the point cloud. Once the scan is complete, the engineer cleans up the point cloud in point cloud modeling software, then converts it to a mesh. They then import the mesh into CAD to create a solid 3D model. In this software, they can adjust the cooling channels from straight to conformal by bending them more closely around the mold cavity.

Once the CAD model is complete, the engineer can convert it to an STL file, then use slicer software to digitally slice the model into the layers that the AM build will create. The engineer then creates a G code program or another type of build file to direct the machine's movements as it builds the mold. Finally, an AM machine, such as a powder bed fusion (PBF) or ultrasonic consolidation machine, can build the part to near-net shape.

Video content available online.

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Reverse Engineering for AM Part Inspection

Even if a manufacturing facility does not need to use RE to create new or replacement parts, it can still be useful for other purposes. If a designer or engineer creates functional prototypes to test a newly designed part, for instance, reverse engineering the part can reveal design issues or potential improvements. For example, after scanning a part, the point cloud may include uneven points that indicate an AM part has printed incorrectly. The designer can then adjust the design file to correct this issue. Alternatively, the point cloud or mesh could indicate a somewhat uneven surface that should be rectified with post-processing, such as sanding or plating.

Similarly, RE can also help manufacturers create improved versions of existing products. Scanning and reverse engineering an existing part can provide designers and engineers a realistic view of the part instead of the original, idealized design file. They can then use the new mesh and CAD files to create a new, improved design. This process can be especially helpful when a manufacturer is first beginning to adopt additive manufacturing. RE is an effective way to import existing part designs and update them for AM processes. AM has greater design flexibility than traditional manufacturing processes, so a shift to AM generally allows designers to improve or enhance a part's features.

A manufacturer may also use RE to maintain an updated inventory of part designs. Scanning existing parts to create a library of CAD files is an effective way to keep records of existing or legacy parts for future reference.

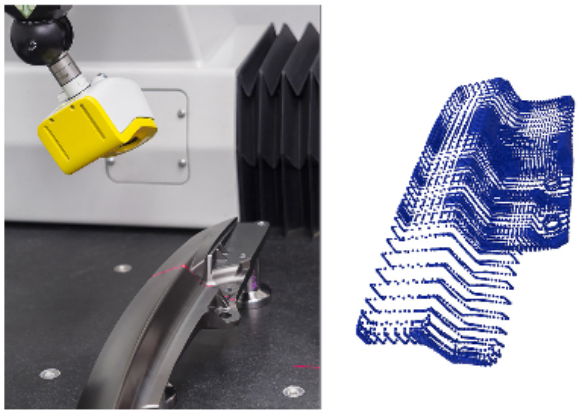

Figure 1: Point clouds from scans can reveal design defects.

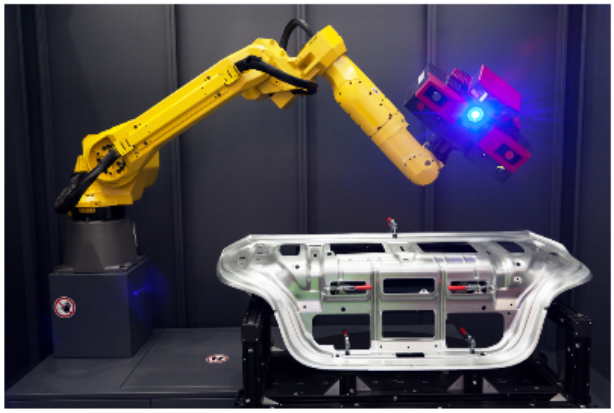

Figure 2: CAD models can begin as part scans.

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Lesson: 10/10

Final Review

- **Noise**: Stray points that are often caused by dust, dirt, or debris.
- **Reverse Engineering**: Can be used to test or prove out existing part designs.
- **STL**: A type of mesh file and mesh software.
- **Polygon**: Created by the interconnected lines of a mesh file.
- **Coordinate Measuring Machine**: Uses a stylus to contact an object and create a point cloud.
- **G Code**: Programming language used to direct AM machine movements.

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Class Vocabulary

3D	Three-dimensional. Having height, width, and depth. 3D models for AM are created with computer-aided design software.
3D Laser Light Scanners	A portable measuring device that uses light and sensors to measure an object's geometric shape. 3D laser light scanners collect large amounts of surface data quickly.
3D Optical Scanning	Capturing the geometric surface data of an object using a combination of light and sensors. 3D optical scanning collects data to be turned into digital 3D models.
Additive Manufacturing	AM. The process of joining or solidifying materials to make an object based on a three-dimensional computer model. Additive manufacturing methods typically build up layers of material to create an object.
AM	Additive manufacturing. The process of joining or solidifying materials to make an object based on a three-dimensional computer model. AM methods typically build up layers of material to create an object.
AMF	Additive Manufacturing File. A newer file format for use in additive manufacturing that contains data for colors, materials, and other part features. The AMF format allows for more complex shapes than the standard STL format, including curved triangles.
Axes	An imaginary straight line or circle used to describe the location or movement of an object in three-dimensional space. The linear axes of the Cartesian coordinate system are the X, Y, and Z axes, and the rotational axes are the A, B, and C axes.
Back Engineering	A process that captures geometric data of an existing object to convert it to a 3D computer-aided design (CAD) model. Back engineering, or reverse engineering, can be used to create an additive manufacturing part from a traditionally manufactured part.
Binder Jetting	An additive manufacturing process in which liquid binder is used to join powdered materials to create parts. Binder jetting can create parts out of metal, plastic, or ceramic.
Build File	Computer information that provides detailed instructions to an additive manufacturing machine. Build files include information about the exact shape and size of each layer.
CAD	Computer-aided design. A computer system used to design a model of a product. CAD models are converted into STL format for use by additive manufacturing machines.
Cartesian Coordinate System	A positioning method that uses three perpendicular linear axes to describe the location of an object in three-dimensional space. The three linear axes in the Cartesian coordinate system are the X axis, Y axis, and Z axis.
CMM Arms	Coordinate measuring machine arms. A portable measuring device consisting of segments connected by joints. CMM arms operate using either contact or noncontact methods.
Computed Tomography	CT. A scanning procedure that utilizes specialized X-ray equipment to generate detailed geometric information about an object. Computed tomography can be used to gather information about the internal features of a part as well as its external features.
Computer-Aided Design	CAD. A computer system used to design a model of a product. Computer-aided design models are converted into STL format for use by additive manufacturing machines.
Conformal Cooling	The use of cooling channel configurations that curve and closely correspond to a part's shape. Conformal cooling provides manufacturers with better heat management and allows them to reduce tool wear and cycle time.
Cooling Channels	A passage through which a cooling fluid is circulated to maintain the desired mold or die temperature. Cooling channels may be straight or contoured.
Coordinate Measuring Machine	CMM. A sophisticated electronic measuring instrument with a flat, polished table that inspects parts in three-dimensional space using either a contact or a noncontact probe. AM manufacturers may use coordinate measuring machines during the reverse engineering process.
Data	A collection of numbers, information, or values about a process or product. Data is transmitted between

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Data A collection of numbers, information, or values about a process or product. Data is transmitted between computers and machines.

G Code A method of programming that pairs address letters with numerical values to form words. G code programs are used in additive manufacturing, CNC machining, and hybrid machining.

Horizontal Parallel to the horizon or ground. A horizontal axis runs side-to-side rather than up-and-down.

Imaging System An assembly of components that records and collects images of an object created during computed tomography. The imaging system is connected to a computer, which uses specialized software to reconstruct a series of two-dimensional (2D) images into a three-dimensional (3D) model.

Laser A device that generates an intense beam of light that can be precisely aimed and controlled. Laser beams are used to measure and capture surface data.

Laser Scanning The process of using a concentrated beam of light to generate data about the geometric shape of an object. Laser scanning is used to create a point cloud that can be turned into a 3D computer-aided design (CAD) model.

Material Extrusion An additive manufacturing method that uses a nozzle to dispense material, usually a thermoplastic filament, onto a build platform. Material extrusion is sometimes referred to as either fused deposition modeling (FDM) or fused filament fabrication (FFF).

Mesh A three-dimensional digital model of an object that consists of a number of interconnected triangles or other shapes. Mesh files resemble a web or netting in the shape of a physical item.

Metrology The science of measurement. Metrology strives for accuracy, precision, and repeatability.

Microchips A minuscule device that conducts electricity and transfers data. Microchips are used in an array of electronics and can be smaller than can be seen with the human eye.

Molds A hollow cavity used to shape material. Molds are filled with liquid material, which takes the shape of the mold as it solidifies.

Near-Net Shape A part that is close to meeting consumer or manufacturer specifications, including tolerance and surface finish, directly after it is manufactured. Near net-shape parts require little subsequent machining.

Noise Unwanted surface data caused by light interference. Noise, or scatter, affects the accuracy of data collection.

Origin A fixed, central point in the Cartesian coordinate system where the three X, Y, and Z axes intersect. The origin has a numerical value of zero.

Photogrammetry A 3D imaging method that digitally combines numerous photographs to create a virtual model of an object. Photogrammetry can be used in reverse engineering, but it is relatively imprecise and inaccurate.

Plating The process of adding a thin layer of metal to serve as a decorative or protective coating on a part. Plating is sometimes used to improve the surface finish of a part.

Point Cloud A large collection of data points of a three-dimensional object. Point clouds are typically used to create computer-aided design (CAD) models of objects.

Point Cloud Modeling Software A computer program or set of operating instructions that allows someone to modify a digital point cloud model. Point cloud modeling software is necessary for most reverse engineering processes.

Polygons A closed shape consisting of line segments that has at least three sides. Triangles, quadrilaterals, rectangles, and squares are all types of polygons.

Post-Processing A finishing process used to clean, grind, or otherwise prepare a manufactured part for delivering to a consumer or other manufacturer. Common additive manufacturing post-processing steps include removing support structures, improving surface finish, and bringing the part into tolerance.

Powder Bed Fusion PBF. An additive manufacturing method that uses thermal energy to either sinter or melt together layers of powdered polymer, metal, ceramic, or other material. Powder bed fusion processes often use either lasers or electron beams as thermal energy sources.

Proprietary Designed by a specific company for use only with its own systems or devices. Proprietary components include physical hardware, such as programming devices, and software components, such as programming software.

Prosthetics An artificial implant, device, or limb that replaces a missing body part. Prosthetics may also refer to the design, fabrication, and fitting process involved with these devices.

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Prototype A preliminary model of a part used to evaluate the look and performance of a design. Prototypes are used to determine the specifications for the final part.

Radiation Energy emitted in the form of particles or waves. Types of radiation include sound, light, and electromagnetic waves.

RE Reverse engineering. A process that captures geometric data of an existing object to convert it to a 3D computer-aided design (CAD) model. RE, or back engineering, can be used to create an additive manufacturing part from a traditionally manufactured part.

Respirators A breathing device worn to prevent inhalation of hazardous substances. Respirators may purify air or supply air to the wearer.

Reverse Engineering RE. A process that captures geometric data of an existing object to convert it to a 3D computer-aided design (CAD) model. Reverse engineering, or back engineering, can be used to create an additive manufacturing part from a traditionally manufactured part.

Scatter Unwanted surface data caused by light interference. Scatter is also known as noise and affects the accuracy of data collection.

Slicer A computer program that divides an STL file into layers for additive manufacturing (AM). Slicers can either be separate computer programs or part of the machine interface.

Software A set of written operating instructions that control a computer. A software program consists of coded formulas and operations that determine the computer's actions.

STL Computer files that represent 3D models as a series of interconnected triangles. STL files are used with additive manufacturing machines.

Structured-Light Scanning Capturing the geometric surface data of an object using a measuring device that projects a pattern of light on the object and scans an image. Structured-light scanning covers a large surface area with each individual scan.

Stylus A precision tip that records measurements. On a CMM, the stylus is the spherical portion of the contact probe that makes contact with the part.

Three-Dimensional 3D. Having height, width, and depth. Three-dimensional models for AM are created with computer-aided design software.

Two-Dimensional 2D. Having length and width, but not depth. Flat shapes are two-dimensional.

Ultrasonic Consolidation UC. A sheet lamination process that uses high-frequency vibrations to bond successive layers of material together to build a final part. Ultrasonic consolidation, or ultrasonic additive manufacturing (UAM), most frequently uses sheets of metal foil or tape as build material.

Vertical Perpendicular to the horizon or ground. A vertical axis runs up-and-down rather than side-to-side.

X-Rays A type of electromagnetic wave with a high frequency and short wavelength. X-rays are used in computed tomography systems to scan the interior of a part and create a digital model for reverse engineering applications.

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